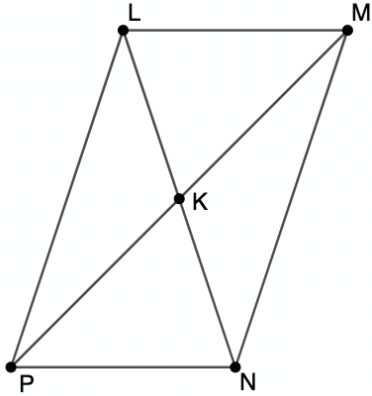


Geometry  
Unit 8  
Lesson 1 Practice

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Parallelogram  $LMNP$  where diagonals  $\overline{LN}$  and  $\overline{MP}$  intersecting at  $K$  as shown.



**Given:**  $LMNP$  is a parallelogram with diagonals  $\overline{LN}$  and  $\overline{MP}$  intersecting at  $K$ .

**Prove:**  $\angle LMN \cong \angle NPL$

Complete the following two-column proof.

| Statement                                                                    | Reason                                                      |
|------------------------------------------------------------------------------|-------------------------------------------------------------|
| $LMNP$ is a parallelogram with diagonals $\overline{LN}$ and $\overline{MP}$ | Given                                                       |
| $\overline{LM} \cong \overline{PN}$ ; $\overline{LP} \cong \overline{MN}$    | 1. Opposite sides of a parallelogram are congruent          |
| 2. $\overline{LN} \cong \overline{LN}$                                       | Reflexive Property of Congruence                            |
| 3. $\triangle LMN \cong \triangle NPL$                                       | 4. SSS $\cong$                                              |
| $\angle LMN \cong \angle NPL$                                                | 5. Corresponding Parts of Congruent Triangles are Congruent |

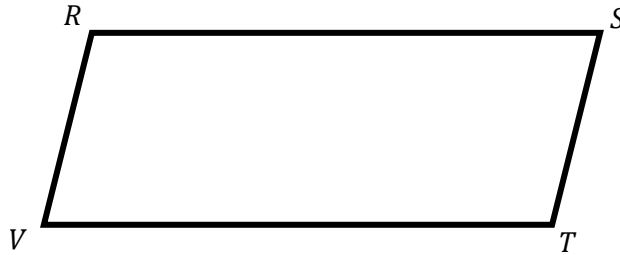
Fill in the blanks with the correct terms to complete the statements.

6. Consecutive interior angles of a parallelogram are supplementary.
7. The diagonals of a parallelogram bisect each other.
8. Opposite angles of a parallelogram are congruent.
9. Opposite sides of a parallelogram are parallel.
10. A quadrilateral containing two set(s) of congruent sides is a parallelogram.

Geometry  
Unit 8  
Lesson 2 Practice

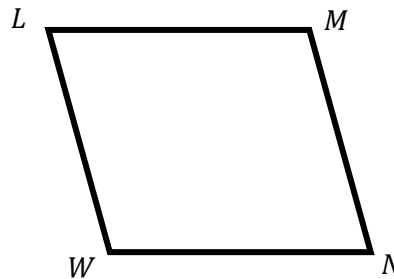
Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Parallelogram  $RSTV$  is shown where  $m\angle S = (4x - 8)^\circ$ ,  $m\angle T = 144^\circ$ , and  $m\angle V = 36^\circ$ .



1. I can conclude ...  
 $m\angle R = 144^\circ$
2. I know this because ...  
Opposite angles in a parallelogram are congruent and have equal measure
3. Since  $RSTV$  is a parallelogram, what is the value of  $x$ ?  
 $x = 11$

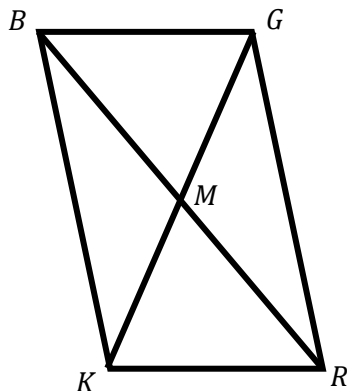
Quadrilateral  $LMNW$  is shown where  $m\angle M = (14x - 8)^\circ$ ,  $m\angle N = (16y - 4)^\circ$ ,  $m\angle W = (5x + 64)^\circ$ , and  $m\angle L = (12y + 16)^\circ$ .



4. What value of  $x$  will make  $LMNW$  a parallelogram?  
 $x = 8$
5. What value of  $y$  will make  $LMNW$  a parallelogram?  
 $y = 5$
6. Given that  $LMNW$  is a parallelogram, what is the  $m\angle M$ ?  
 $m\angle M = 104^\circ$

Quadrilateral  $BGRK$  is shown with diagonals  $\overline{BR}$  and  $\overline{GK}$  that intersect at point  $M$ .

$MB = 8z - 6$ ,  $MR = 32$ ,  $MG = 28$ , and  $MK = 3t + 3$ , where  $t = 8$ .



Charlene and Jordan are discussing whether  $BGRK$  is a parallelogram. Charlene says  $BGRK$  is a parallelogram and Jordan says  $BGRK$  is not a parallelogram.

7. Who is correct?

Jordan is correct. Diagonals of a parallelogram bisect each other which means that  $KM = MG$ . Using the given information  $KM = 3(8) + 3 = 27$  and  $MG = 28$ . Since  $KM \neq MG$ , the quadrilateral  $BGRK$  is not a parallelogram.

8. Which property of parallelograms was used to determine if  $BGRK$  is a parallelogram?

Diagonals of a parallelogram bisect each other

9. Given that  $BGRK$  is a parallelogram, what is the value of  $z$ ?

It is not possible to determine the value of  $z$  because  $BGRK$  is not a parallelogram so there is not enough information to identify relationships with lengths of sides and diagonals of this quadrilateral.

10. What is the length of  $\overline{MK}$ ?

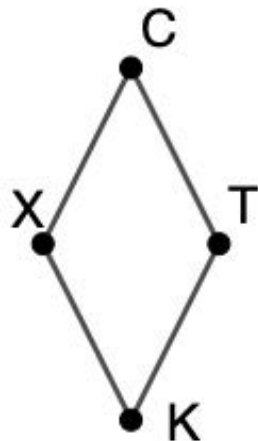
$$KM = 3(8) + 3 = 27$$

Geometry  
Unit 8  
Lesson 3 Practice

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Rhombus  $CTKX$  is shown where  $CT = 15x - 3$  and  $TK = 7x + 1$ .

1. Select all true properties of  $CTKX$ .
  - ☐ Opposite sides are parallel.
  - ☐ Opposite sides are congruent.
  - ☐ Consecutive angles are congruent.
  - ☐ All angles are right angles.
  - ☐ Diagonals are perpendicular.
  - ☐ All sides are congruent.



2. What is the value of  $x$ ?

$$x = \frac{1}{2}$$

3. What is the length of  $\overline{CT}$ ?

$$CT = 4.5$$

Rectangle  $TRXP$  is shown with diagonals  $\overline{TX}$  and  $\overline{RP}$  which intersect at point  $Z$ .

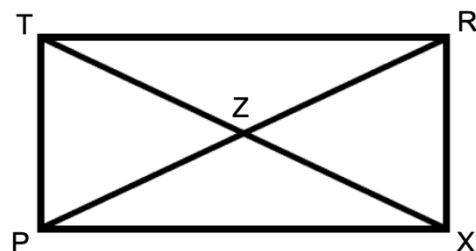
4. Write two congruent statements using the diagonals of  $TRXP$ .

$$\overline{TX} \cong \overline{RP}$$

$$\overline{TZ} \cong \overline{XZ}$$

$$\overline{PZ} \cong \overline{RZ}$$

$$\overline{TZ} \cong \overline{XZ} \cong \overline{PZ} \cong \overline{RZ}$$



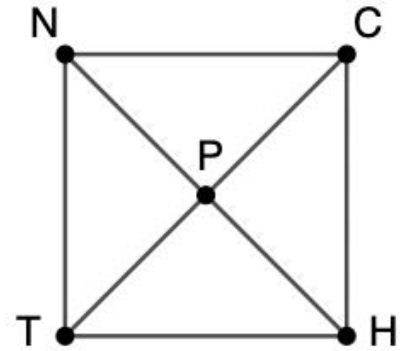
5.  $TZ = (5y - 8)$ , and  $ZX = (3y + 16)$ . What is the value of  $y$ ?

$$y = 11$$

6. What is the length of  $\overline{RP}$ ?

$$RP = 98$$

Quadrilateral  $NCHT$  is shown with diagonals  $\overline{NH}$  and  $\overline{CT}$  intersecting at point  $P$ .



7. What value of  $x$  makes  $NCHT$  a square, if  $m\angle TNC = 90^\circ$ ,  $m\angle NCH = 90^\circ$ ,  $CH = 21$ , and  $HT = (4x - 7)$ ?

$$x = 7$$

8. Given  $NCHT$  is a square with  $PC = (5y - 2)$  and  $PT = (3y + 4.74)$ . What is the length of  $\overline{NH}$ ?

$$y = 3.37$$

$$NH = 29.7$$

9. What are some similarities between  $CTKX$  and  $NCHT$ ?

Opposite sides are parallel

Opposite sides are congruent

Opposite angles are congruent

Consecutive angles are supplementary

Diagonals are perpendicular to each other

Diagonals bisect each other

All sides are congruent

10. What are some differences between  $TRXP$  and  $NCHT$ ?

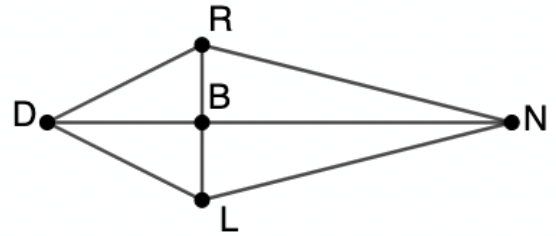
In  $NCHT$  diagonals are perpendicular to each other

In  $NCHT$  all four sides are congruent

Geometry  
Unit 8  
Lesson 4 Practice

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Kite  $RNLD$  is shown with diagonals  $\overline{ND}$  and  $\overline{RL}$  intersecting at point  $B$  where  $BD = 24$  and  $RB = 15$ .



1. What is the length of  $\overline{RD}$  ?

$$RD = \sqrt{24^2 + 15^2} = \sqrt{801} \approx 28.3$$

2. Given  $NB = 12$ . What is the length of  $\overline{NR}$  ?

$$NR = \sqrt{12^2 + 15^2} = \sqrt{369} \approx 19.21$$

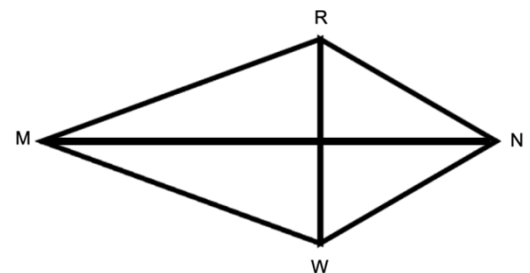
3. What is the perimeter of  $RNLD$ ?

$$\text{Perimeter} = 2\sqrt{801} + 2\sqrt{369} = 2(28.3) + 2(19.21) \approx 95.02$$

$MRNW$  is a kite where  $m\angle RMN = (5x - 17)^\circ$ ,  $m\angle MRN = (11x + 13)^\circ$ ,  $m\angle WNM = (3y - 2)^\circ$ , and  $m\angle MWN = 112^\circ$ .

4. What is the value of  $x$ ? Round your answer to the nearest hundredth.

$$x = 9$$



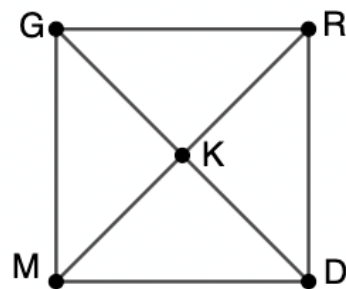
5. What is the value of  $y$ ? Round your answer to the nearest hundredth.

$$y = 14$$

6. What is the measure of  $\angle RNW$ ?

$$m\angle RNW = 80^\circ$$

Quadrilateral  $GRDM$  is shown with diagonals  $\overline{GD}$  and  $\overline{MR}$  intersecting at point  $K$ .



7. What must be true for  $GRDM$  to be a rectangle?

Answers may vary. For  $GRDM$  to be a rectangle,

$$\overline{GR} \cong \overline{MD} \text{ and } \overline{GM} \cong \overline{RD}$$

OR

$$\overline{GD} \cong \overline{MR}$$

8. Given  $DK = 9z$ , and  $DM = 18$ . What value of  $z$  will make  $GRDM$  a square?

$$z = \sqrt{2} \approx 1.41$$

9. Given  $GRDM$  is a square, what is the length of  $\overline{RM}$ ?

$$RM = MK + KR = 9\sqrt{2} + 9\sqrt{2} = 18\sqrt{2} \approx 25.46$$

10. Given  $GRDM$  is a square, what is the length of  $\overline{GM}$ ?

$$GM = DM = 18$$

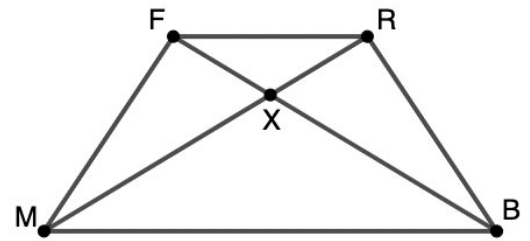


Geometry  
Unit 8  
Lesson 5 Practice

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Given:  $FRBM$  is an isosceles trapezoid,  $\angle MFX \cong \angle BRX$ .

Prove:  $\triangle MXB$  is an isosceles triangle.



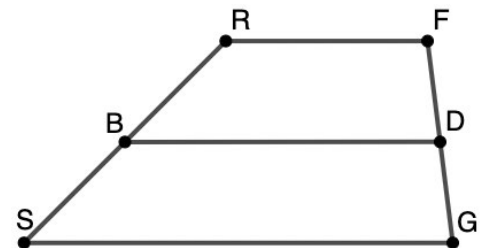
| Statement                                | Reason                                                        |
|------------------------------------------|---------------------------------------------------------------|
| $FRBM$ is an isosceles trapezoid         | Given                                                         |
| $\angle FMB \cong \angle RBM$            | 1. <b>Base angles of an isosceles trapezoid are congruent</b> |
| $\angle MFX \cong \angle BRX$            | Given                                                         |
| 2. $\overline{MB} \cong \overline{MB}$   | Reflexive Property of Congruence                              |
| 3. $\triangle FMB \cong \triangle RBM$   | AAS Congruence                                                |
| 4. $\angle FBM \cong \angle RMB$         | Corresponding Parts of Congruent Triangles are Congruent      |
| $\triangle MBX$ is an isosceles triangle | 5. <b>Base angles of an isosceles triangle are congruent</b>  |

Trapezoid  $SRFG$  is shown.  $\overline{BD}$  is a midsegment of  $SRFG$ .  
 $FD = 4x + 34$ ,  $DG = 8x - 14$ ,  $RB = 8y + 41$ ,  $SB = 13y - 4$ .

6. Find the values of  $x$  and  $y$ .

$$x = 12$$

$$y = 9$$

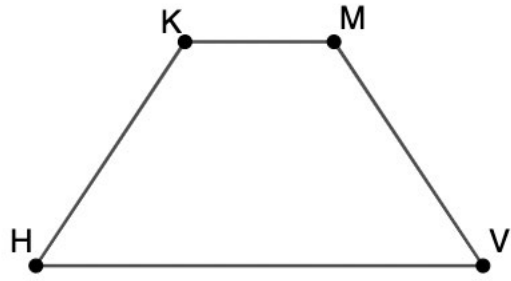


7. Find the lengths of  $\overline{RS}$  and  $\overline{DG}$ .

$$RS = 226$$

$$DG = 82$$

Isosceles trapezoid  $KMVH$  is shown where  $m\angle KHV = (16z - 6)^\circ$ ,  $m\angle HKM = (21z + 38)^\circ$ ,  $m\angle KMV = (16t - 26)^\circ$ .



8. What is the value of  $z$ ?

$$z = 4$$

9. What is the value of  $t$ ?

$$t = 6$$

10. What are  $m\angle VHK$ ,  $m\angle HKM$ ,  $m\angle KMV$ , and  $m\angle MVH$ ?

$$m\angle VHK = 58^\circ$$

$$m\angle MVH = 58^\circ$$

$$m\angle HKM = 122^\circ$$

$$m\angle KMV = 122^\circ$$

Geometry  
Unit 8  
Lesson 6 Practice

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Two congruent angles of an isosceles trapezoid have measures  $(8x + 14)^\circ$  and  $(12x - 8)^\circ$ .

1. What is the value of  $x$ ?

$$\begin{aligned}8x + 14 &= 12x - 8 \\4x &= 22 \\x &= \frac{11}{2}\end{aligned}$$

2. What are the measures of each angle in the trapezoid?

$$\begin{aligned}8\left(\frac{11}{2}\right) + 14 &= 58^\circ \\180 - 58 &= 122^\circ\end{aligned}$$

For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral  $GRMW$  has two pairs of congruent sides and consecutive angles are supplementary.

- ☒ Parallelogram
- ☒ Rectangle
- ☒ Square
- ☒ Rhombus
- ☐ Kite

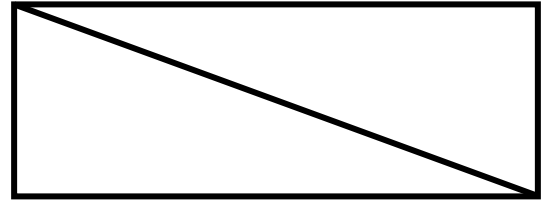
4. Quadrilateral  $BNTV$  has four congruent sides and diagonals are perpendicular to each other.

- ☐ Parallelogram
- ☐ Rectangle
- ☒ Square
- ☒ Rhombus
- ☐ Kite

5. Quadrilateral  $KPSX$  has two pairs of congruent sides and one pair of opposite congruent angles.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☒ Kite

Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of  $32x$  feet, width of  $46x$  feet and the diagonal of the building is 200 feet.



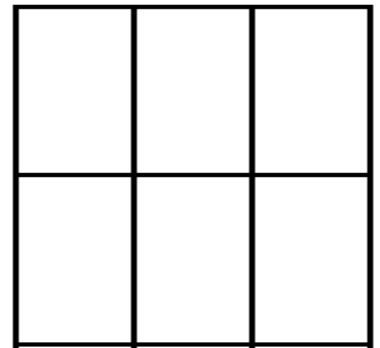
6. What is the value of  $x$ ? Round answers to the nearest hundredth.

$$\begin{aligned}(32x)^2 + (46x)^2 &= 200^2 \\ 3140x^2 &= 40000 \\ x^2 &= 12.739 \\ x &\approx 3.5692\end{aligned}$$

7. What is the height of the building? Round answers to the nearest hundredth.  
*height  $\approx 32(3.5692) \approx 114.21$  ft*

8. What is the width of the building? Round answers to the nearest hundredth.  
*width  $\approx 46(3.5692) \approx 164.18$  ft*

A wall paneling with a total area of 18 square feet is made up of 6 rectangles, as shown. There are two congruent rectangles along the height and three across the width. Each rectangle in the paneling is  $(3x + 1)$  feet in height and  $(2x - 1)$  feet in width.



9. What is the value of  $x$ ? Round your answer to the nearest hundredth.

$$\begin{aligned}18 &= 6[(2x - 1)(3x + 1)] \\ x &\approx 0.904\end{aligned}$$

10. What are the dimensions of the portion of the wall covered by the paneling?

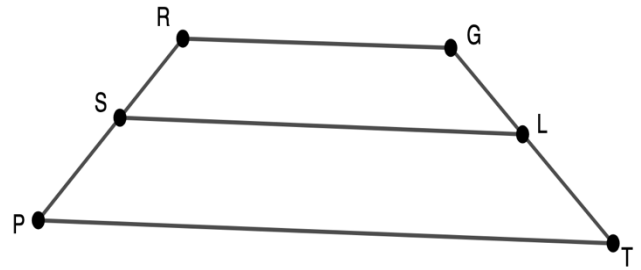
$$\begin{aligned}\text{height of 1 panel} &= 3(0.904) + 1 \\ \text{height of 2 panels} &= 2[3(0.904) + 1] \\ \text{height} &\approx 7.424 \text{ ft}\end{aligned}$$

$$\begin{aligned}\text{width of 1 panel} &= 2(0.904) - 1 \\ \text{width of 3 panels} &= 3[2(0.904) - 1] \\ \text{width} &\approx 2.424 \text{ ft}\end{aligned}$$

Geometry  
Unit 8  
Lesson 7 Practice

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

$PTGR$  is a trapezoid as shown where  $\overline{PT} \parallel \overline{RG}$ .  $\overline{SL}$  is the midsegment of  $PTGR$  with  $SL = 18$ ,  $PT = 9x - 5$ , and  $RG = 7x - 3$ .



1. Write an equation to show the relationship between  $\overline{SL}$  and the bases of the trapezoid.

$$SL = \frac{RG + PT}{2}$$

2. What is the value of  $x$ ? Round your answer to the nearest hundredth.

$$18 = \frac{7x - 3 + 9x - 5}{2}$$

$$36 = 7x - 3 + 9x - 5$$

$$44 = 16x$$

$$x = \frac{11}{4} = 3.75$$

3. What are the lengths of  $\overline{PT}$  and  $\overline{RG}$ ? Round your answers to the nearest hundredth.

$$PT = 9\left(\frac{11}{4}\right) - 5 = 19.75$$

$$RG = 7\left(\frac{11}{4}\right) - 3 = 16.25$$

Trapezoid  $HKMV$  is shown with midsegment  $\overline{DP}$  where  $VH = 9x + 12$ ,  $MK = 6x$ , and  $DP = 10x$ .



4. What is the value of  $x$ ? Round your answer to the nearest tenth.

$$10x = \frac{9x + 12 + 6x}{2}$$

$$20x = 9x + 12 + 6x$$

$$5x = 12$$

$$x = 2.4$$

5. What is the length of  $\overline{DP}$ ?

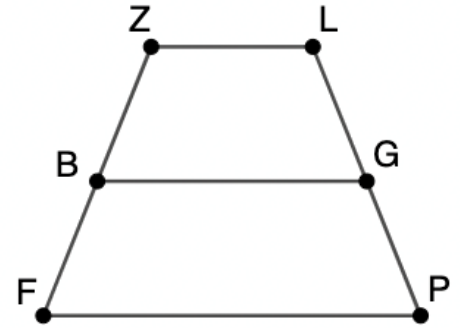
$$DP = 10(2.4) = 24$$

6. What are the lengths of  $\overline{VH}$  and  $\overline{MK}$ ? Round your answer to the nearest tenth.

$$VH = 9(2.4) + 12 = 33.6$$

$$MK = 6(2.4) = 14.4$$

Isosceles trapezoid  $FPLZ$  is shown with midsegment  $\overline{BG}$  where  $ZL = 5x + 1$ ,  $FP = 8x + 2$ ,  $BG = 4x + 4$ ,  $BZ = 3.5$ .



7. What is the value of  $x$ ?

$$4x + 4 = \frac{5x + 1 + 8x + 2}{2}$$

$$8x + 8 = 5x + 1 + 8x + 2$$

$$5x = 5$$

$$x = 1$$

8. What is the length of  $\overline{BG}$ ?

$$BG = 4(1) + 4 = 8$$

9. What are the lengths of  $\overline{ZL}$  and  $\overline{FP}$ ?

$$ZL = 5(1) + 1 = 6$$

$$FP = 8(1) + 2 = 10$$

10. What is the perimeter of  $FPLZ$ ?

$$\text{Perimeter} = ZL + LP + FP + FZ$$

$$\text{Perimeter} = 6 + 2(3.5) + 10 + 2(3.5)$$

$$\text{Perimeter} = 30$$